Leakage-Aware, Explainable Intrusion Detection for Industrial Control Systems Using Sentence Embeddings and MITRE ATT&CK for ICS

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***Abstract*—Industrial control systems (ICS) and the Industrial Internet of Things (IIoT) are becoming increasingly connected, which also exposes them to a wider range of cyber threats. Recent research has explored the use of large language models (LLMs) and MITRE ATT&CK for ICS to detect and explain intrusions. However, an earlier study introduced this approach as a concept and left prototype development and testing for future work. In this paper, we implement and evaluate the approach. We compare four supervised models—logistic regression, random forest, LightGBM and a multilayer perceptron—with a zero-shot ATT&CK similarity method across five public datasets: BATADAL, Edge-IIoTset, X-IIoTID, TON_IoT-Network and TON_IoT-Modbus. All datasets are processed using a leakage-aware protocol that removes identifiers and counters, eliminates duplicates after identifier removal, and uses stratified or chronological data splits. De-duplication removes 99.7% of Edge-IIoTset and 71.5% of TON_IoT-Modbus records. On TON_IoT-Modbus, LightGBM accuracy falls from 0.97 to 0.54 after duplicates are removed. On the three network datasets that retain learnable patterns, LightGBM achieves attack F1 scores of 0.995, 0.996 and 0.998. BATADAL and TON_IoT-Modbus do not provide a reliable learnable signal under this protocol. The zero-shot method also performs poorly on real traffic, with ROC-AUC values of 0.43–0.51 on four datasets and 0.74 on TON_IoT-Network. We combine the three learnable datasets by converting each event into text, encoding it with a sentence encoder and classifying the resulting embeddings with LightGBM. The merged model achieves a pooled held-out F1 score of 0.986 and provides approximate ATT&CK technique matches as explanations. However, its ROC-AUC is only 0.55 on handwritten attack descriptions. We deploy the system as a local web application that keeps structured-event detection separate from an experimental free-text mode.**

***Index Terms*—industrial control systems, IIoT, intrusion detection, sentence embeddings, MITRE ATT&CK for ICS, data leakage, LightGBM, zero-shot detection**

I. Introduction

Industrial control systems (ICS), including Supervisory Control and Data Acquisition (SCADA) systems, support essential services such as water distribution, power grids and manufacturing. As these systems become connected to enterprise networks and the Internet through the Industrial Internet of Things (IIoT), their exposure to cyber threats increases [1]. Signature-based intrusion detection systems (IDS) look for known indicators of compromise and may miss attacks that have not been observed before. Machine-learning (ML) based IDS can instead learn patterns in system behaviour and may generalise to previously unseen attacks [1], [2].

Attacks such as Stuxnet (2010), BlackEnergy (2016), Triton (2017) and EKANS (2020) show that cyber incidents involving ICS can have physical consequences, including damaged centrifuges, blackouts and plant shutdowns [3]. Conventional detection often relies on signature fields such as IP addresses, ports and MAC addresses, which an attacker can change. In contrast, the tactics and techniques in MITRE ATT&CK for ICS describe attacker behaviour [3], [4]. Ann *et al.* therefore proposed an ICS intrusion detection approach in which a large language model (LLM) is fine-tuned using information from ATT&CK for ICS and real attack cases [3]. Their study presents the approach as a concept and leaves prototype development and testing for future work. As a result, it remains unclear whether the proposed idea can detect attacks in real ICS and IIoT traffic.

Evaluating such a system raises two important challenges. First, near-perfect results reported on public intrusion detection datasets can be inflated by duplicated records and identifying columns. These can allow a model to memorise a testbed instead of learning patterns associated with attack behaviour [10], [11]. Second, public ICS and IIoT datasets represent different types of observations: an hourly plant measurement, a packet or a connection flow. Their original columns therefore cannot simply be combined into one dataset.

This paper makes four contributions:

1) We apply the same leakage-aware evaluation protocol to five public datasets. It removes identifiers, deduplicates records after identifier removal, checks individual features for leakage, runs ablation tests and compares results with a version that does not remove duplicates.
2) We compare four supervised models with a zero-shot ATT&CK similarity method. Performance is measured using accuracy, precision, recall, F1, ROC-AUC and false-positive rate on held-out test sets.
3) We build a merged detector by converting each event into text, encoding it with a sentence encoder and clas-

sifying the embeddings with LightGBM. Cross-dataset and merge experiments are used to determine which datasets can be combined.

4) We develop a local web application that returns a detection verdict, attack probability and approximate ATT&CK technique matches. The application keeps structured-event detection separate from the experimental free-text mode and documents the latter's limitations.

The remainder of this paper is organised as follows. Section II introduces the background and the earlier study's pipeline; Section III reviews related work; Section IV describes the datasets; Section V explains the methodology; Section VI presents the results; Section VII discusses limitations; and Section VIII concludes the paper.

II. Background

A. MITRE ATT&CK for ICS

MITRE ATT&CK is a knowledge base that documents adversary behaviour observed in real-world attacks. It is organised into tactics, which describe an adversary's objective (the "why"), and techniques, which describe how that objective is achieved. The ATT&CK matrix connects these tactics and techniques [3], [4].

The ICS matrix described in the earlier study includes 12 tactics, such as Initial Access, Execution, Persistence and Impact, and refers to 81 techniques. The snapshot used in our experiments contains 97 technique entries, each with a name, an associated tactic and a description. Real incidents can be described in terms of the techniques used during the attack; Stuxnet and PLC-Blaster are examples discussed in [3]. Since these techniques focus on behaviour rather than network addresses or ports, an ATT&CK-based detector is less dependent on fields that an attacker can easily change.

B. The Base Study's Pipeline and Our Adaptation

The earlier study proposes a three-stage pipeline [3]. The first stage collects tactics, techniques and procedures (TTPs) from real attacks using ATT&CK, other open sources such as GitHub, Google and CISA, and additional heuristics. The second stage preprocesses the collected information by mapping each real attack to the ATT&CK for ICS matrix and storing the result in a structured file. In the third stage, a pre-trained large language model—ChatGPT, Bard and LLaMA are mentioned as examples—is fine-tuned so that defenders can ask about likely attacker techniques and possible next steps. The study presents this as a proposed approach and states that prototype development and testing will be carried out in future work; it does not report experimental results.

Our work follows the same three stages but adds the implementation and evaluation that were missing from the earlier study. For the first two stages, we build a structured reference set from the ATT&CK for ICS knowledge base. It contains 97 technique descriptions and 271 procedure examples, which describe how real attacks use particular techniques.

For the modelling stage, we do not fine-tune a generative LLM. Instead, we use a pre-trained sentence-embedding model to compare event descriptions with ATT&CK technique texts. We also train supervised classifiers on public ICS and IIoT datasets to test whether the approach can detect attacks in real traffic. This design retains the ATT&CK grounding of the earlier study while keeping the system lightweight enough to run locally and evaluate against labelled data.

III. Related Work

Machine-learning methods for unknown-attack detection. Zhang *et al.* used zero-shot learning with a sparse autoencoder to create a semantic representation of attack characteristics [2]. Li *et al.* combined random-forest feature selection with spatial clustering to support zero-shot detection through attribute representations [12]. Hindy *et al.* explored autoencoders for outlier-based zero-day detection [13]. Kumar and Sinha used heavy-hitter and graph-based techniques to detect high-volume zero-day attacks [14], while Sabeel *et al.* proposed a two-stage detector based on network flows [15].

Hwang *et al.* and Abdalgawad *et al.* applied autoencoders and generative adversarial networks to network traffic and IoT-23 data [16], [17]. For industrial environments, Mokhtari *et al.* used machine learning to identify anomalies in ICS measurement data [18].

Hybrid detectors. HMS-IDS combines a stacked ensemble of decision tree, naive Bayes, *k*-nearest neighbours and logistic regression models to detect known attacks. It then uses a cluster-labelled *k*-means stage to handle unknown attacks. The system is evaluated on CIC-ToN-IoT, KDD-99, NSL-KDD and CICIDS2017 [1]. Our work follows a similar staged approach and also uses preprocessing, feature engineering, multiple base learners and evaluation across several datasets. However, we replace the original tiers with embedding-based and supervised components designed for ICS data.

Language models for cybersecurity and ATT&CK. Ann *et al.* describe an ICS intrusion detection approach that fine-tunes an LLM using ATT&CK for ICS information [3]. Other studies apply language models to security tasks. Ferrag *et al.* combine Edge-IIoTset with a BERT-based model for cyber-threat detection in IoT networks [5]. Ali and Kostakos combine machine-learning-based anomaly detection with GPT-3.5 to support explainable threat detection in IT infrastructure [6]. Garza *et al.* examine whether LLMs contain knowledge of ATT&CK threat behaviour and mitigations [7]. Broader surveys review LLM security and privacy, as well as machine-learning approaches to IoT security [8], [9].

Sentence encoders such as Sentence-BERT and MiniLM produce compact text embeddings that can be used for similarity search [19], [20]. In this work, we use them both to compare events with ATT&CK technique descriptions in the zero-shot setting and to create a shared representation across datasets.

Dataset quality. Engelen *et al.* identified labelling and dataset-construction problems in CICIDS2017 that can affect reported performance [10]. Arp *et al.* discussed sampling bias, data snooping and other recurring pitfalls in machine-learning security research [11]. Our evaluation addresses one of these

concerns directly by checking for and removing duplicate-driven leakage.

Table I summarises representative intrusion-detection studies. The performance figures are those reported by the original authors, as summarised in [1]. Since the studies use different datasets and evaluation protocols, their results are not directly comparable with ours.

IV. Datasets

We use five public datasets that represent different types of industrial and IoT data, including process telemetry, packet-level traffic, connection flows and register polling (Table II).

BATADAL [21] contains hourly SCADA measurements from a water distribution network. Its fields include tank levels, pump and valve flows and statuses, and junction pressures. Dataset 03 represents a normal year, while dataset 04 contains five labelled attack windows. Rows marked `−999` are unlabelled and are treated as normal in our experiments, although some may contain unlabelled attacks.

Edge-IIoTset [22] is a packet-level dataset covering protocols such as Modbus, MQTT, HTTP and DNS. It includes 14 attack types and contains 2,219,201 rows in its original form. After identifier removal and de-duplication, 6,513 unique rows remain. The DDoS_UDP and MITM source files contain misaligned columns, which we account for during field selection. The dataset contains records with misaligned fields; these were aligned with the corresponding columns during field selection.

X-IIoTID [23] combines network-flow features with host telemetry, including CPU and memory measurements and OSSEC alerts. It covers 18 attack types and represents activities such as data exfiltration, command-and-control, tampering and lateral movement.

TON_IoT-Network [24] provides Zeek connection logs with connection, DNS, HTTP and SSL fields. It includes nine attack types, and 77.0% of its records are attack traffic after de-duplication.

TON_IoT-Modbus [24] contains four Modbus register values per polling interval and covers five attack types.

V. Proposed Methodology

A. System Overview

The workflow is illustrated in Fig. 1. Raw datasets first pass through a leakage-aware preprocessing stage. The resulting cleaned records feed two parallel branches. The first branch trains four supervised models on each dataset's native features and reports performance separately for each dataset. In the second branch, each event is converted into a short text description and embedded into a shared vector space. These embeddings support both the zero-shot ATT&CK matcher and the merged LightGBM detector. The detector and matcher are then connected to a web application, which provides a classification verdict, an attack probability and approximate ATT&CK technique matches.

B. Leakage-Aware Preprocessing

The same preprocessing policy is applied consistently across all datasets:

- **Identifier removal.** IP addresses, timestamps, dates, packet sequence and acknowledgement numbers, checksums, stream counters and raw option bytes are removed because they can identify a capture session rather than the underlying behaviour. The two original port fields are replaced with a single `service_port` value representing the lower, server-side port. Raw strings—including payloads, URIs, queries, DNS names, certificates and user-agent strings—are replaced by their lengths and counts of characters commonly associated with injection attempts (such as apostrophes, quotation marks, angle brackets, percent signs and semicolons). The HTTP method and first 80 characters of the URI are retained only for text descriptions; they are excluded from tabular model features, and host names are stripped from URIs.
- **De-duplication after identifier removal.** Records that differ only in an identifier are treated as duplicates. De-duplication is performed after identifier fields have been removed, preventing behaviourally identical feature vectors from being split between training and test sets. Records whose identical feature vectors carry conflicting labels are counted and reported: 61 in Edge-IIoTset, 6 in X-IIoTID and 19 in TON_IoT-Network.
- **Sampling.** After de-duplication, each dataset is limited to at most 250,000 records. Sampling is stratified by the finest available attack-type label, retaining at least 1,000 records per class or all available records when a class contains fewer than 1,000. Only X-IIoTID requires subsampling.
- **Data splits.** Data is divided into training, validation and test sets using a 70/15/15 split stratified by attack type. BATADAL is the exception: it is split chronologically, with training records dated before 2016-10-20, validation records before 2016-11-15, and later records assigned to testing. Attack classes with fewer than 10 unique records after de-duplication are kept in the training set only. BATADAL is also evaluated using leave-one-attack-window-out (LOWO) validation.

C. Supervised Models

Four models are trained on the native features of each dataset: logistic regression as the baseline, random forest [25], LightGBM [26] and a multilayer perceptron implemented in PyTorch [27]. The first two models use scikit-learn [28]. Class weights are set to "balanced", and no oversampling is applied. A limited hyperparameter search is evaluated only on the validation set, using attack F1 for binary classification and macro-F1 for multiclass classification. The multiclass models reuse the configuration selected for the binary task. The decision threshold is 0.5, and the random seed is 42.

For binary classification, logistic regression estimates $p(y{=}1 \mid \mathbf{x}) = \sigma(\mathbf{w}^{\top}\mathbf{x} + w_0)$, where $\sigma(z) = 1/(1 + e^{-z})$.

TABLE I
COMPARISON OF RELATED INTRUSION DETECTION STUDIES (METRICS AS REPORTED BY THE ORIGINAL AUTHORS; NOT DIRECTLY COMPARABLE).

Study	Year	Key concept	Dataset	Reported performance	Zero-day
Zhang *et al.* [2]	2020	Zero-shot learning with sparse autoencoder	NSL-KDD	Acc. 88.3%	Yes
Li *et al.* [12]	2019	Random forest and spatial clustering	NSL-KDD	Acc. 77.28%	Yes
Hwang *et al.* [16]	2020	Unsupervised CNN/autoencoder	USTC-TFC2016, Mirai	Acc. 99.77%, F1 99.55%	Yes
Abdalgawad *et al.* [17]	2022	AAE, BiGAN, KNN, RF	IoT-23	Acc. 97%, F1 0.94	Yes
Kumar and Sinha [14]	2021	Heavy-hitter and graph technique	Real-time attack data	Acc. 91.62%, F1 0.95	Yes
Saurabh *et al.* (HMS-IDS) [1]	2025	Stacked ensemble and k-means	CIC-ToN-IoT	Acc. 99.49% (known), 98.936% (unknown)	Yes
Ann *et al.* [3]	2024	LLM fine-tuned on ATT&CK for ICS data	— (concept)	No evaluation reported	—
This work	2026	Leakage-aware embeddings, LightGBM and ATT&CK matching	5 ICS/IIoT datasets	F1 0.986 (merged, held-out)	Not shown

TABLE II
DATASET OVERVIEW AFTER IDENTIFIER REMOVAL AND DE-DUPLICATION (SEED 42).

Dataset	Data type	Raw rows	Unique rows	Dup. (%)	Attack types	Attack (%)	Train/val/test	Test (attack %)
BATADAL	SCADA telemetry, hourly	12,938	12,938	0.0	1	1.7	11,353/624/961	961 (8.3)
Edge-IIoTset	Packet-level traffic	2,219,201	6,513	99.7	14	58.7	4,561/976/976	976 (58.6)
X-IIoTID	Flow and host data	820,834	734,479	10.5	18	44.7	174,995/37,499/37,499	37,499 (46.9)
TON_IoT-Network	Zeek connection flows	211,043	104,722	50.4	9	77.0	73,305/15,708/15,709	15,709 (77.0)
TON_IoT-Modbus	Modbus register values	31,106	8,868	71.5	5	57.7	6,207/1,330/1,331	1,331 (57.7)

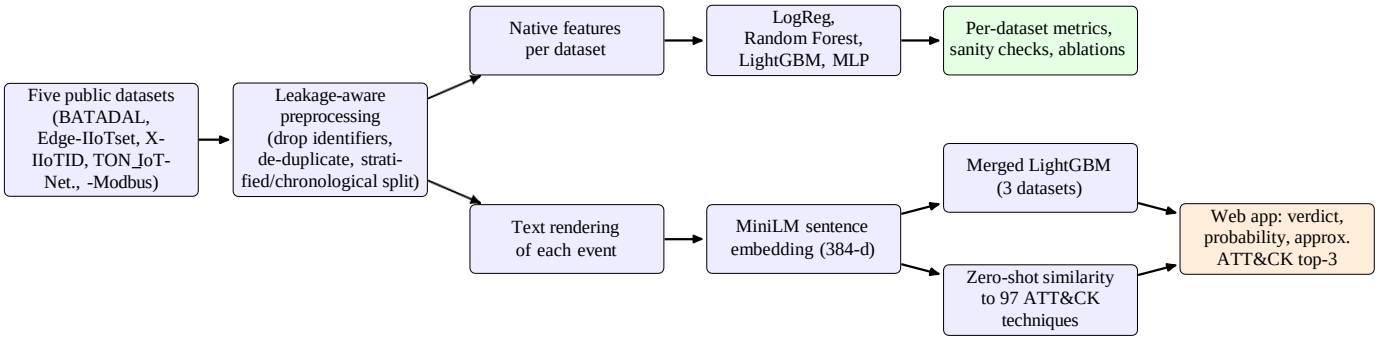


Fig. 1. Proposed leakage-aware, embedding-based intrusion detection pipeline.

LightGBM forms an additive ensemble of gradient-boosted trees, $F(\mathbf{x}) = \sum_m f_m(\mathbf{x})$, and produces the probability $p = \sigma(F(\mathbf{x}))$.

D. Text Rendering and Embeddings

To make datasets with different feature columns comparable, each record is converted into a concise event description using the retained fields available in that dataset. Depending on the source, these fields can include the service, protocol, connection state, byte and packet counts, HTTP method and URI prefix. The resulting description is encoded using the all-MiniLM-L6-v2 sentence encoder, which is based on a six-layer MiniLM architecture. The sentence encoder converts each event description into a 384-dimensional vector [19], [20]. These embeddings are cached so that every experiment uses the same vector representations.

E. Zero-Shot ATT&CK Matching

Our first approach follows the base study's idea of grounding intrusion detection in MITRE ATT&CK for ICS knowledge [3], without training or fine-tuning a generative model. Let $\mathbf{e}_x$ denote the embedding of an event description, and let $\mathbf{e}_{t_j}$ denote the embedding of the text describing ATT&CK for ICS technique t_j, where j ranges from 1 to 97. The similarity score and predicted label are defined as follows:

$$s(x) = \max_j \cos(\mathbf{e}_x, \mathbf{e}_{t_j}), \qquad \hat{y} = \mathbb{1}[s(x) \geq \tau]. \quad (1)$$

For deployment, the threshold is set to $\tau = 0.4158$, calibrated using the project's technique-description examples. This score represents cosine similarity; it should not be interpreted as a calibrated probability. We also evaluate a variant in which the threshold is tuned using a training sample.

Algorithm 1 Leakage-aware training and merged detection

function TRAIN(datasets $D_1, \ldots, D_K$)
  for each D_k **do**
    $D_k \leftarrow$ DROPIDENTIFIERS(D_k)
    $D_k \leftarrow$ DEDUPLICATE(D_k)
    $(\mathrm{Tr}_k, \mathrm{Va}_k, \mathrm{Te}_k) \leftarrow$ SPLIT(D_k)
    $E_k \leftarrow$ EMBED(RENDERTEXT(Tr_k))
  end for
  $M \leftarrow$ LIGHTGBM($\bigcup_k E_k$, equal weight per dataset)
  return M
end function
function DETECT(event x, model M)
  $\mathbf{e} \leftarrow$ EMBED(RENDERTEXT(x))
  $p \leftarrow M(\mathbf{e})$; $\hat{y} \leftarrow \mathbb{1}[p \geq 0.5]$
  $T \leftarrow$ top-3 techniques by cosine similarity to $\mathbf{e}$
  return $(\hat{y}, p, T)$
end function

F. Merged Detector

The merged approach trains a single LightGBM classifier, with 31 leaves, on the 384-dimensional embeddings from the datasets selected in the experiments described in Section VI. Each dataset is given equal weight, and the decision threshold is fixed at 0.5. A model card records the classifier, its training procedure, per-dataset test metrics and the single-dataset baselines evaluated on the same test records.

For each prediction, the system returns a classification verdict and an attack probability. It also lists the three ATT&CK techniques whose embeddings have the highest similarity to the event embedding. These technique matches are presented as approximate explanations only; they do not contribute to the classifier's decision.

G. Web Application

The web application is built with FastAPI and provides two analysis modes. In structured-event mode (`/api/analyze_event`), the submitted event fields are converted into the same template used during training and passed to the merged detector. Events that do not contain any recognised fields are rejected.

The second mode accepts free text (`/api/analyze`). This mode is experimental and has lower confidence: it uses the zero-shot similarity method, displays a warning banner and also reports the merged model's attack probability for reference.

A sample-data loader provides 60 events from the held-out test splits, with 20 events drawn from each of the three datasets used for the merged evaluation. After analysis, the loader reveals the true label, allowing users to compare the system's prediction with the ground-truth result.

H. Computational Cost

Among the supervised baselines, the multilayer perceptron is the most expensive model to train. Text-based processing also adds a fixed embedding cost for each row. For a tree ensemble with T trees of depth d, inference requires $O(Td)$ time per row. In the merged detector, most inference time is spent in the sentence encoder's transformer pass over the event description; the LightGBM prediction on the resulting 384-dimensional vector is comparatively inexpensive. Because embeddings are computed once and cached for training and evaluation, repeated experiments are less computationally expensive.

I. Evaluation Metrics

Let TP, TN, FP and FN denote true positives, true negatives, false positives and false negatives, respectively. We treat attacks as the positive class and report the following metrics:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}; \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{FPR} = \frac{FP}{FP + TN}; \quad \text{FNR} = \frac{FN}{FN + TP}$$

We also report ROC-AUC and precision-recall AUC, along with macro-F1 for multiclass results. Since accuracy can be misleading on imbalanced test sets, F1 and ROC-AUC are treated as the primary metrics.

VI. Results and Observations

A. Per-Dataset Supervised Results

Table III reports binary attack-detection results on each held-out test set, and Fig. 2 summarises the attack-class F1 scores. The tree ensembles achieve the best or tied-best F1 on Edge-IIoTset, X-IIoTID and TON_IoT-Network, with scores of 0.995, 0.996 and 0.998, respectively. LightGBM obtains an ROC-AUC of 1.000 on all three datasets.

BATADAL produces a different outcome: the tree models do not identify any attacks (recall = 0.000). Their accuracy of 0.915 mainly reflects the proportion of normal records, since attacks make up only 8.3% of the test set. On TON_IoT-Modbus, all models perform close to chance, with accuracy between 0.49 and 0.54 and ROC-AUC between 0.48 and 0.54.

The multiclass results in Table IV follow the same overall ranking. LightGBM reaches macro-F1 scores of 0.994 on X-IIoTID and 0.959 on TON_IoT-Network. Edge-IIoTset has a lower macro-F1 of 0.599 because some attack classes contain very few unique test records, while others have no unique test records after de-duplication. Consequently, the per-class scores are unstable.

B. BATADAL Leave-One-Window-Out Validation

Because the chronological BATADAL split contains few attacks, we also evaluate each of its five labelled attack windows separately. The mean attack F1 is no higher than 0.199 for any model: logistic regression achieves 0.199 ± 0.203, LightGBM 0.131 ± 0.131, the MLP 0.087 ± 0.078 and random forest 0.021 ± 0.041. The best mean ROC-AUC is 0.789 ± 0.166, obtained by random forest.

TABLE III
BINARY ATTACK DETECTION ON HELD-OUT TEST SETS (ACC: ACCURACY, P: PRECISION, R: RECALL, F1 FOR THE ATTACK CLASS).

Dataset	Logistic Regression				Random Forest				LightGBM				MLP			
	Acc	P	R	F1	Acc	P	R	F1	Acc	P	R	F1	Acc	P	R	F1
BATADAL	.828	.011	.013	.012	.915	.000	.000	.000	.915	.000	.000	.000	.742	.038	.087	.053
Edge-IIoTset	.964	.989	.949	.969	.992	.993	.993	.993	**.994**	**.996**	.993	**.995**	.968	.991	.955	.972
X-IIoTID	.972	.984	.955	.969	.994	**.998**	.990	.994	**.996**	.997	**.994**	**.996**	.986	.991	.979	.985
TON_IoT-Network	.976	.991	.978	.985	**.998**	**.998**	**.999**	**.998**	**.998**	**.999**	.998	**.998**	.991	.996	.992	.994
TON IoT-Modbus	.526	.604	.517	.557	.531	.571	.753	.649	.541	.610	.568	.588	.485	.554	.552	.553

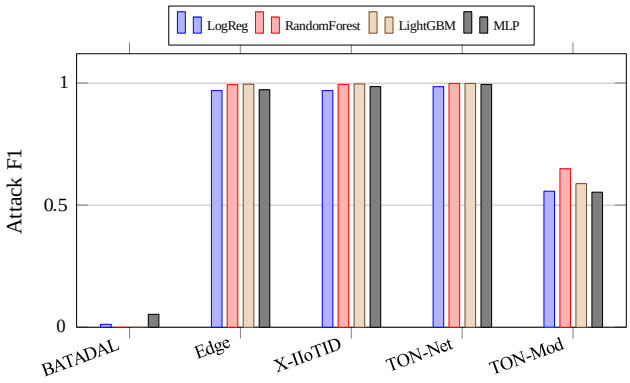


Fig. 2. Attack F1 of the four supervised models on each held-out test set.

TABLE IV
MULTICLASS RESULTS ON HELD-OUT TEST SETS (ACCURACY / MACRO-F1).

Dataset	LogReg	RF	LightGBM	MLP
Edge-IIoTset	.930/.551	.955/.533	**.962/.599**	.666/.401
X-IIoTID	.964/.938	.994/.990	**.996/.994**	.969/.939
TON_IoT-Net.	.779/.687	.977/.959	**.981/.959**	.813/.758
TON_IoT-Modbus	.092/.084	.369/.119	**.249/.168**	.128/.096

On the chronological test split, logistic regression and the MLP achieve ROC-AUC values of only 0.095 and 0.167, respectively. This indicates that their scores are reversed across attack windows. Under our evaluation protocol, BATADAL therefore does not provide a reliable supervised detection signal.

C. Leakage and Sanity Checks

Table V summarises the leakage checks. For each dataset, we identify the feature with the highest AUC on the training set, remove it, retrain LightGBM and count the test feature vectors that also occur in the training data. No single feature explains the strong results. The leading features are `http.request.method` for Edge-IIoTset (AUC 0.865), `dst_pkts` for TON_IoT-Network (0.837) and `service_port` for X-IIoTID (0.783). Removing each feature leaves F1 essentially unchanged: 0.995 to 0.996, 0.996 to 0.996 and 0.998 to 0.998, respectively. Only 0.0–2.8% of test feature vectors also occur in the training set.

TABLE V
SINGLE-FEATURE LEAK CHECKS (LIGHTGBM).

Dataset	Top feature	AUC	F1 w/o it	Seen in train
BATADAL	`L_T1`	.744	.000	0 (0.0%)
Edge-IIoTset	`http.request.method`	.865	.996	27 (2.8%)
X-IIoTID	`service_port`	.783	.996	0 (0.0%)
TON_IoT-Net.	`dst_pkts`	.837	.998	33 (0.2%)
TON_IoT-Modbus	`FC2_Read_Discrete_Value`	.512	.666	0 (0.0%)

TABLE VI
LIGHTGBM WITHOUT AND WITH DE-DUPLICATION.

	No de-dup.			De-dup.	
Dataset	Acc	F1	Seen in train	Acc	F1
Edge-IIoTset	.9960	.9936	99.1%	.9939	.9947
X-IIoTID	.9966	.9967	8.9%	.9962	.9959
TON_IoT-Net.	.9988	.9992	53.7%	.9975	.9984
TON_IoT-Modbus	.9715	.9722	93.6%	.5409	.5880

D. Effect of De-duplication

Table VI compares LightGBM performance with and without de-duplication. Without de-duplication, many test records have feature vectors that also appear in the training set: 99.1% for Edge-IIoTset, 53.7% for TON_IoT-Network and 93.6% for TON_IoT-Modbus.

Removing duplicates has little effect on Edge-IIoTset and TON_IoT-Network, suggesting that the signal in these datasets remains after repeated records are removed. TON_IoT-Modbus behaves differently: accuracy falls from 0.9715 to 0.5409, while F1 decreases from 0.9722 to 0.5880. This substantial drop indicates that the earlier high score was largely driven by duplicate records rather than reliable detection. We therefore exclude TON_IoT-Modbus from the merged dataset.

E. Zero-Shot ATT&CK Matching

The zero-shot method does not provide reliable attack detection on these datasets (Table VII). It labels 0% of records as attacks in BATADAL, X-IIoTID and TON_IoT-Modbus. In contrast, it flags 83.4% of Edge-IIoTset records and 100% of TON_IoT-Network records as attacks. The resulting F1 scores of 0.651 and 0.870 are driven by the method labelling nearly every record as an attack; the corresponding false-positive rates are 0.899 and 0.998.

The method's ROC-AUC is between 0.43 and 0.51 on four datasets, including BATADAL and X-IIoTID, indicating

TABLE VII
ZERO-SHOT ATT&CK MATCHING AT THE DEPLOYED THRESHOLD (0.416).

Dataset	Rows flagged	F1	ROC-AUC	Top-1 agrees
BATADAL	0.0%	.000	.477	0.0%
Edge-IIoTset	83.4%	.651	.434	0.0%
X-IIoTID	0.0%	.000	.459	0.4%
TON_IoT-Net.	100.0%	.870	.740	0.0%
TON_IoT-Modbus	0.0%	.000	.509	0.0%

TABLE VIII
CROSS-DATASET ATTACK F1 (ROC-AUC) OF AN EMBEDDING PROBE. ROWS: TRAINING SET; COLUMNS: TEST SET.

Train \ Test	BATADAL	Edge	X-IIoTID	TON-Net	TON-Mod
BATADAL	.125 (.49)	.118 (.77)	.002 (.49)	.000 (.48)	.000 (.50)
Edge-IIoTset	.154 (.58)	**.995** (1.00)	.627 (.55)	.891 (.85)	.000 (.51)
X-IIoTID	.154 (.59)	.280 (.27)	**.873** (.93)	.880 (.57)	.732 (.51)
TON_IoT-Net.	.000 (.53)	.708 (.93)	.659 (.52)	**.928** (.98)	.000 (.51)
TON_IoT-Mod.	.154 (.50)	.739 (.50)	.640 (.50)	.870 (.50)	.732 (.50)

TABLE IX
MERGED-GROUP ATTACK F1 VERSUS SINGLE-DATASET F1 (EMBEDDING REPRESENTATION).

Merge group	Tested on	Merged	Single	Δ
Edge + TON-Net	Edge	.992	.995	−.003
Edge + TON-Net	TON-Net	.932	.928	+.004
Edge + X-IIoTID	X-IIoTID	.818	.873	−.055
X-IIoTID + TON-Net	X-IIoTID	.764	.873	−.109
X-IIoTID + TON-Net	TON-Net	.941	.928	+.013
Network trio	Edge	.991	.995	−.004
Network trio	X-IIoTID	.739	.873	−.134
Network trio	TON-Net	.937	.928	+.008
All five	X-IIoTID	.828	.873	−.045
All five	TON-Net	.969	.928	+.041
All five	BATADAL	.000	.125	−.125

performance close to or below chance. TON_IoT-Network is an exception, with an ROC-AUC of 0.74. The top-ranked ATT&CK technique agrees with the analyst mapping for only 0–0.4% of records in BATADAL.

The zero-shot method also performs poorly on the base study's procedure and data-description benchmark. Across 97 ATT&CK techniques, its top-1 and top-3 mapping accuracies are 0.321 and 0.480. For attack-versus-benign classification, it reaches an accuracy of 0.745, correctly classifying 208 of 271 attack descriptions and 8 of 13 legitimate sentences. However, the threshold was selected using these same sentences to maximise performance. The results may therefore overstate how well the method would perform on new descriptions in a real-world setting.

F. Cross-Dataset Generalisation

Table VIII presents a cross-dataset evaluation using a class-weighted logistic-regression probe. The probe is trained on 5,000 embedded records from one dataset and evaluated on 2,000 records from another; for BATADAL, all 961 available records are used. Same-dataset performance is strongest for Edge-IIoTset (F1 = 0.995), TON_IoT-Network (F1 = 0.928) and X-IIoTID (F1 = 0.873).

Edge-IIoTset and TON_IoT-Network show transferability in both directions: training on Edge-IIoTset and testing on TON_IoT-Network gives F1 = 0.891 and AUC = 0.85, while the reverse direction reaches AUC = 0.93. By contrast, X-IIoTID transfers poorly to Edge-IIoTset (AUC = 0.27) and TON_IoT-Network (AUC = 0.57).

Some cross-dataset results show high F1 alongside an AUC close to 0.50. This can happen when a model predicts nearly every record as an attack; such results should not be interpreted as successful generalisation. This behaviour is visible, for example, in the TON_IoT-Modbus training row. Using the ten flow features shared by X-IIoTID and TON_IoT-Network, LightGBM trained on X-IIoTID achieves F1 = 0.805 (AUC = 0.82) on TON_IoT-Network. In the reverse direction, F1 falls to 0.548 (AUC = 0.55).

G. Merge Experiments and Dataset Selection

Table IX compares models trained on merged dataset groups with the corresponding single-dataset models, using the same embedding representation. Combining Edge-IIoTset with TON_IoT-Network has almost no effect: Edge-IIoTset changes by −0.003 F1, while TON_IoT-Network improves by +0.004. TON_IoT-Network is the only dataset that clearly improves in the larger combinations, gaining +0.041 when all five datasets are merged.

When X-IIoTID is included in a merge, its attack F1 decreases by 0.045 to 0.134, although its ROC-AUC remains at or above 0.91. A row-level merge of X-IIoTID and TON_IoT-Network using their shared flow features does not reduce performance: F1 remains 0.985 for X-IIoTID and 0.997 for TON_IoT-Network, matching their single-dataset results. BATADAL does not benefit from merging, and TON_IoT-Modbus is excluded because of the duplicate-related results discussed earlier.

Based on these findings, the final merged dataset combines Edge-IIoTset, TON_IoT-Network and X-IIoTID. X-IIoTID is retained despite its reduction in F1 because it contributes attack scenarios less represented in the other datasets, including data exfiltration, command-and-control, tampering, false-data injection, Modbus register reading and lateral movement. BATADAL and TON_IoT-Modbus are excluded.

H. Final Merged Model

The final model, `ics_ids_merged_v1`, is a LightGBM classifier trained on MiniLM embeddings from the three selected datasets. Model selection uses the validation splits only. The held-out test records in Table X are not used for model selection, early stopping or threshold selection.

The pooled held-out F1 score is 0.986 (ROC-AUC 0.998; false-positive rate 0.020). A LightGBM model trained on a single dataset using its native columns scores 0.3 to 2.8 F1 points higher on the same test rows. The largest difference occurs on X-IIoTID (0.996 for the single-dataset model versus 0.968 for the merged model). Thus, merging provides one

TABLE X
MERGED MODEL ON HELD-OUT TEST SETS (THRESHOLD 0.5).

Test set	Rows	F1	ROC-AUC	FPR
Edge-IIoTset	976	.993	.999	.010
TON_IoT-Network	2,000	.994	1.000	.013
X-IIoTID	2,004	.968	.995	.027
Pooled	4,980	**.986**	**.998**	**.020**

model and one text representation, at the cost of a small reduction in performance for some attack sources.

I. Web Application Evaluation

We submitted 60 held-out test events through the application API, with 20 events from each of the three datasets. The merged detector classified 58 of the 60 events correctly: 20 of 20 from Edge-IIoTset, 20 of 20 from TON_IoT-Network, and 18 of 20 from X-IIoTID. The two misclassified events were a man-in-the-middle attack, assigned an attack probability of 0.088, and a fuzzing attack, assigned a probability of 0.038; both were missed. The website was checked for syntax and serving functionality only. No user study was conducted.

J. Domain Shift

The merged model is trained on template text generated from structured fields. When evaluated on the 271 handwritten attack descriptions and 15 benign sentences in the base study's procedure benchmark, it classified 62.4% of the attack descriptions and 66.7% of the benign sentences as attacks (ROC-AUC 0.553), a result close to chance. A manually written, normal-looking HTTP flow was also classified as an attack with high probability (0.974). Because 77% of the TON_IoT-Network flows are attacks, the model may have learned an association between HTTP traffic and attack activity. Accordingly, the application uses the merged model only for structured-event detection and presents free-text analysis as experimental.

VII. DISCUSSION AND LIMITATIONS

In-testbed scores. F1 scores above 0.99 are specific to the individual testbeds. Although the results pass the leakage checks, cross-dataset transfer is poor; therefore, they should not be interpreted as evidence of equivalent accuracy in a real industrial plant.

Small unique data. After de-duplication, Edge-IIoTset contains only 6,513 unique rows, including a 976-row test set. Its scores and per-class results are therefore subject to substantial uncertainty, and several attack classes have too few unique examples for reliable evaluation.

Dataset balance. TON_IoT-Network contains 77% attack traffic, while Edge-IIoTset contains 58.6%. Models may exploit these class priors; accordingly, FPR and ROC-AUC should be considered alongside F1.

Approximate explanations. The listed ATT&CK techniques are the closest matches based on embedding similarity and do not affect the classification verdict. The mapping from dataset attack labels to ATT&CK techniques was created by the authors and requires expert review before it can be treated as ground truth. Two mapped techniques—Spoof Reporting Message and Unauthorized Command Message—are absent from the 97-technique reference set, and the reference entries do not include technique identifiers.

No zero-day claim. The merged model was not evaluated on unseen attack classes, for example through leave-one-attack-type-out testing. Therefore, no claim is made about zero-day detection. This differs from the zero-day focus of the hybrid detector in [1].

Limited tuning and single seed. The hyperparameter search was limited in scope, and all results use a single random seed (42). Small performance differences, such as those between Random Forest and LightGBM on TON_IoT-Network, should therefore not be treated as statistically significant.

BATADAL. Some attacks in BATADAL dataset 04 are deliberately unlabelled, and rows with a value of `-999` are treated as normal. This labelling noise may conceal part of the underlying signal.

VIII. CONCLUSION AND FUTURE WORK

Following the work in [3], we implemented the proposed concept and evaluated it under a protocol designed to account for several forms of information leakage across five open-source ICS and IIoT datasets.

De-duplication substantially changed the main findings. It removed most of the usable Edge-IIoTset rows and showed that TON_IoT-Modbus was effectively unlearnable. LightGBM was the best-performing or tied-best model on each learnable dataset, achieving F1 scores of 0.995, 0.996 and 0.998 on Edge-IIoTset, X-IIoTID and TON_IoT-Network, respectively. In contrast, the zero-shot ATT&CK similarity approach failed to detect attacks in real traffic.

Merging Edge-IIoTset, TON_IoT-Network and X-IIoTID using text embeddings produced a single detector with a pooled held-out F1 score of 0.986. It provided approximate ATT&CK-style explanations with a small performance cost relative to the single-dataset models, while performing close to chance on handwritten descriptions.

The accompanying web application keeps structured-event detection separate from experimental free-text analysis.

Future work will include evaluating the approach on additional ICS testbeds and using leave-one-attack-type-out splits to assess its ability to detect previously unseen attacks. Further directions include developing the encoder, or training a compact language model on ATT&CK-referenced descriptions, to establish a shared representation for free text and structured events; obtaining expert validation of the ATT&CK mapping; calibrating attack scores so that they provide reliable probability estimates; repeating experiments with multiple random seeds and reporting confidence intervals; and conducting a user study to evaluate the usefulness of the explanations.

DATA AND CODE AVAILABILITY

All datasets are public [21]–[24]. The web application, the study scripts and the full report are available at

https://github.com/rhythm452/icsids-llm-attack-classifier. Trained model files are kept outside that repository.

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